Problem. Consider the limit $\lim_{x\to 2}(-3x+2)$.

(a) What is the value of this limit?

(b) Write down the ε, δ definition in this case.

(c) Suppose you are given $\varepsilon = 1/2$. What is a choice of δ that works?

Solution. (a) We can evaluate this limit by plugging in the corresponding x value, as linear functions are continuous:

$$\lim_{x\to 2}(-3x+2) = -3(2)+2 = -4$$

(b) For any $\varepsilon > 0$, there is a $\delta > 0$ such that if $0 < |x-2| < \delta$, then $|(-3x+2)+4| < \varepsilon$.

(c) In general, for a linear function with slope $a \neq 0$, given ε we can take $\delta = \varepsilon/|a|$. In this case we get $\delta = 1/6$. To see that this works, note that

$$|(-3x+2)+4| = |-3x+6| = |(-3)(x-2)| = 3|x-2|$$

Thus if $0 < |x-2| < 1/6$, $|(-3x+2)+4| = 3|x-2| < 1/2$.

Problem. Differentiate the following function.

$$f(x) = \frac{1}{1+\sqrt{2x^2+1}}$$

Solution. The easiest way to differentiate this is to apply the chain rule twice. We can write $f(x) = g(h(x))$, where $g(x) = 1/x$ and $h(x) = 1+\sqrt{2x^2+1}$. First, we can apply the chain rule to differentiate h:

$$h'(x) = \frac{4x}{2\sqrt{2x^2+1}} = \frac{2x}{\sqrt{2x^2+1}}$$

Now applying the chain rule to f, we have

$$f'(x) = g'(h(x))h'(x) = \frac{-1}{(1+\sqrt{2x^2+1})^2}\cdot\frac{2x}{\sqrt{2x^2+1}} = \frac{-2x}{(1+\sqrt{2x^2+1})^2\sqrt{2x^2+1}}$$

Problem. Evaluate the following definite integral:

$$\int_{-\pi/2}^{\pi/2} \frac{\cos(x)}{\sqrt{2+2\sin(x)}}\mathrm{d}x$$

The problem as stated above contains an error on my part; at $x = -\pi/2$, the function is not continuous, as the denominator becomes 0. It turns out that the resulting integral still has a sensible finite value, but this is outside the scope of what we have seen so far. Below, I've written up a solution of a similar integration problem.

Problem (Corrected)**.** Evaluate the following definite integral:

$$\int_{-\pi/2}^{\pi/2} \frac{\cos(x)}{\sqrt{3+2\sin(x)}}\mathrm{d}x$$

Solution. We use the substitution $u = 3 + 2\sin(x)$. We then have $\mathrm{d}u = 2\cos(x)\mathrm{d}x$, and so $\cos(x)\mathrm{d}x = \frac{1}{2}\mathrm{d}u$. Plugging in $x = -\pi/2$, we get $u = 1$, as $\sin(-\pi/2) = -1$, and plugging in $x = \pi/2$, we get $u = 5$, as $\sin(\pi/2) = 1$. The integral can thus be rewritten as

$$\int_1^5 \frac{1}{2\sqrt{u}}\mathrm{d}u = \int_1^5 \frac{1}{2}u^{-1/2}\mathrm{d}u = \left[u^{1/2}\right]\Big|_1^5 = \sqrt{5} - 1$$